1. Find the values of a and b if the function $f(x)=\begin{cases} -x^2 & x<0 \\ ax+b & 0\le x<1 \\ \sqrt{x+3} & x\ge 1 \end{cases}$ is continuous.

$$\lim_{x\to 0^-}\left(-x^2\right)=\lim_{x\to 0^+}\left(ax+b\right)=f\left(0\right)$$
$$0=b$$
$$\lim_{x\to 1^-}\left(ax+b\right)=\lim_{x\to 1^+}\left(\sqrt{x+3}\right)=f\left(1\right)$$
$$a+b=2\to \text{since } b=0\ ,a=2$$

2. Let $f(x)=\dfrac{x^2-9}{x}$. State the discontinuities of $f(x)$ and find their type.

At x=0, infinite discontinuity.

3. A designer is experimenting with a cylindrical can with a fixed height of 15 cm. find the rate of change of volume with respect to radius when the radius is 4 cm. the volume of a cylinder is $V=\pi r^2 h$.

$$\begin{aligned} V'\left(r\right) &= \lim_{h\to 0}\frac{V\left(r+h\right)-V\left(r\right)}{h} \\ &= \lim_{h\to 0}\frac{\pi\left(r+h\right)^2 H-\pi r^2 H}{h} \\ &= \lim_{h\to 0}\frac{\pi H\left[\left(r+h\right)^2-r^2\right]}{h} \\ &= \pi H\lim_{h\to 0}\frac{\left[\cancel{r^2}+2rh+h^2-\cancel{r^2}\right]}{h} \\ &= \pi H\lim_{h\to 0}\frac{\cancel{h}\left(2r+h\right)}{\cancel{h}} \\ &= 2\pi rH \end{aligned}$$

$$at\ \ H=15cm\ \ and\ \ r=4cm: V'\left(4\right)=2\pi\left(4cm\right)\left(15cm\right)=120\pi cm^2$$

4. Let $f\left(x\right)=ax^2+bx+c$. Find a,b and c so that the tangent to the graph $y=f\left(x\right)$ has slope 16 where x $x=2$ and has x-intercepts (0,0) and (8,0).

$$f'\left(x\right)=2ax+b$$
$$f'\left(2\right)=16\to 2a\left(2\right)+b=16\to 4a+b=16 \quad (1)$$
$$f\left(0\right)=0\to c=0$$
$$f\left(8\right)=0\to 64a+8b=0\to 8a+b=0 \quad (2)$$
$$\begin{cases} 4a+b=16 \\ 8a+b=0 \end{cases}\Rightarrow a=-4\ \&\ \ b=32$$

5. Is the following statement true? Justify your answer.

It is possible for a limit to exist even though the function may not be continuous at the point of question.

Yes it's true. For example: $f(x)=\begin{cases} x & if \quad x\neq 0 \\ 1 & if \quad x=0 \end{cases}$ ***is not continuous at x=0, however***

$\lim_{x\to 0} f(x)=\lim_{x\to 0}(x)=0$.

6. Function $f(x)=\begin{cases} \dfrac{x^2-1}{x+1} & if \quad x\neq -1 \\ -2k+1 & if \quad x=-1 \end{cases}$ is continuous at $x=-1$.Find the value of k.

$$\lim_{x\to -1}\left(\frac{x^2-1}{x+1}\right)=-2k+1$$

$$\lim_{x\to -1}\frac{(x-1)\cancel{(x+1)}}{\cancel{x+1}}=-2k+1$$

$$\lim_{x\to -1}(x-1)=-2k+1$$

$$-2=-2k+1\to k=\frac{3}{2}$$

7. At what point on the parabola $y=3x^2$ is the slope of the tangent equal to 24?

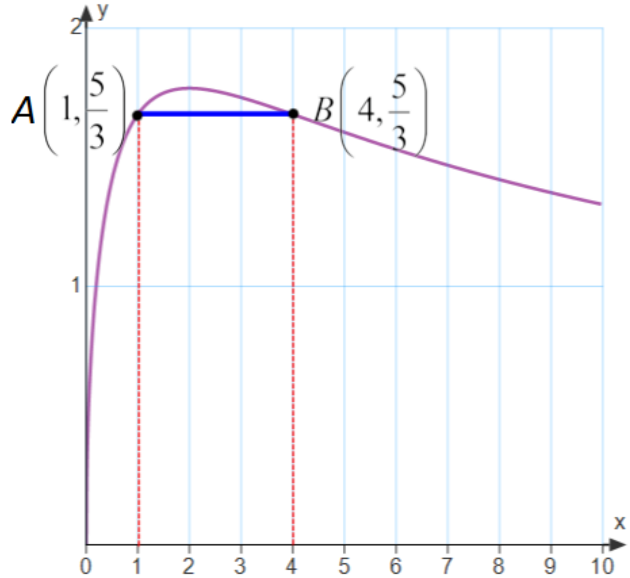
$$y=3x^2\to y'=6x$$
$$6x=24\to x=4$$
$$\therefore (4,48)$$

8. Find the points on the curve $y=1-\dfrac{1}{x}$ where the tangent line is perpendicular to the line $y=1-4x$.

$$y=1-\frac{1}{x}\to y'=\frac{1}{x^2} \qquad \Rightarrow \frac{1}{x^2}=\frac{1}{4}\to x^2=4\to x=\pm 2$$

$$y=1-4x\to m=-4\to m_{\perp}=\frac{1}{4} \qquad \therefore A\left(2,\frac{1}{2}\right),\ B\left(-2,\frac{3}{2}\right)$$

9. Find the average rate of change of the function $f(x)=\frac{5\sqrt{x}}{x+2}$ between $x=1$ and $x=4$. What is the instantaneous rate of change at x= 1.



$$\underset{[1,4]}{ARoC}=\frac{f(4)-f(1)}{4-1}$$
$$=\frac{\frac{10}{6}-\frac{5}{3}}{3}$$
$$=0$$

$$IRoC=\lim_{h\to0}\frac{f(1+h)-f(1)}{h}$$
$$=\lim_{h\to0}\frac{\frac{5\sqrt{1+h}}{1+h+2}-\frac{5}{3}}{h}$$
$$=\lim_{h\to0}\frac{\frac{5\sqrt{1+h}}{h+3}-\frac{5}{3}}{h}$$
$$=5\lim_{h\to0}\frac{3\sqrt{1+h}-(h+3)}{3h(h+3)}\times\frac{3\sqrt{1+h}+(h+3)}{3\sqrt{1+h}+(h+3)}$$
$$=5\lim_{h\to0}\frac{9(1+h)-(h+3)^2}{3h(h+3)\left[3\sqrt{1+h}+(h+3)\right]}$$
$$=5\lim_{h\to0}\frac{\cancel{9}+9h-h^2-6h\cancel{-9}}{3h(h+3)[6]}$$
$$=5\lim_{h\to0}\frac{3h-h^2}{3h(h+3)[6]}$$
$$=5\lim_{h\to0}\frac{\cancel{h}(3-h)}{18\cancel{h}(h+3)}$$
$$=\frac{5}{18}$$

10. Evaluate the following limits, if they exist. Show your work.

a) $\lim_{x\to\frac{2}{3}y}\frac{27x^3-8y^3}{9x^2-4y^2}=\lim_{x\to\frac{2}{3}y}\frac{\cancel{(3x-2y)}\left(9x^2+6xy+4y^2\right)}{\cancel{(3x-2y)}(3x+2y)}=3y$

b) $\lim_{x\to\infty}7^{-(x-3)}=7^{-\infty}=0$

c) $\lim\limits_{x\to 0} 5^{x-3} = 5^{-3} = \dfrac{1}{125}$

d) $\lim\limits_{x\to 2} \dfrac{x^3 - x^2 - 4x + 4}{x^2 - 4} = \lim\limits_{x\to 2} \dfrac{\cancel{(x-2)}\left(x^2 + x - 2\right)}{\cancel{(x-2)}(x+2)} = 1$

e) $\lim\limits_{x\to 0} \dfrac{\sqrt{4+x+x^2} - 2}{x} \times \dfrac{\sqrt{4+x+x^2} + 2}{\sqrt{4+x+x^2} + 2}$

$= \lim\limits_{x\to 0} \dfrac{\cancel{4} + x + x^2 \cancel{-4}}{x\left(\sqrt{4+x+x^2} + 2\right)}$

$= \lim\limits_{x\to 0} \dfrac{\cancel{x}(1+x)}{\cancel{x}\left(\sqrt{4+x+x^2} + 2\right)}$

$= \dfrac{1}{4}$

f) $\lim\limits_{x\to\infty} \dfrac{\cancel{\left(\frac{2}{3}\right)^x}^{\,0} - 2}{\cancel{\left(\frac{2}{3}\right)^x}_{\,0} + 2} = \dfrac{-2}{2} = -1$

$g)$ $\lim\limits_{x\to 4^+} \dfrac{x-4}{\sqrt{x} - 2} \times \dfrac{\sqrt{x} + 2}{\sqrt{x} + 2}$

$= \lim\limits_{x\to 4^+} \dfrac{\cancel{(x-4)}\left(\sqrt{x} + 2\right)}{\cancel{x-4}}$

$= 4$

h) $\lim\limits_{x\to 2} \dfrac{\sqrt{x^2+5} - \sqrt{x+7}}{x-2} \times \dfrac{\sqrt{x^2+5} + \sqrt{x+7}}{\sqrt{x^2+5} + \sqrt{x+7}}$

$= \lim\limits_{x\to 2} \dfrac{\left(x^2+5\right) - (x+7)}{(x-2)\left[\sqrt{x^2+5} + \sqrt{x+7}\right]}$

$= \lim\limits_{x\to 2} \dfrac{x^2 - x - 2}{(x-2)\left[\sqrt{x^2+5} + \sqrt{x+7}\right]}$

$= \lim\limits_{x\to 2} \dfrac{\cancel{(x-2)}(x+1)}{\cancel{(x-2)}\left[\sqrt{x^2+5} + \sqrt{x+7}\right]}$

$= \dfrac{1}{2}$

i) $\lim_{x\to 2}\frac{\sqrt[3]{2x-5}+1}{x-2}$

$Let\ \sqrt[3]{2x-5}=u$

$2x-5=u^3$

$x=\frac{u^3+5}{2}$

$$\lim_{x\to 2}\frac{\sqrt[3]{2x-5}+1}{x-2}=\lim_{u\to -1}\frac{u+1}{\frac{u^3+5}{2}-2}$$

$$=\lim_{u\to -1}\frac{2(u+1)}{u^3+5-4}$$

$$=\lim_{u\to -1}\frac{2(u+1)}{u^3+1}$$

$$=\lim_{u\to -1}\frac{2\cancel{(u+1)}}{\cancel{(u+1)}(u^2-u+1)}$$

$$=\frac{2}{3}$$

j) $\lim_{x\to\infty}\frac{(2x+3)^2}{5-2x-5x^2}$

$$=\lim_{x\to\infty}\frac{4x^2+12x+9}{-5x^2-2x+5}$$

$$=\lim_{x\to\infty}\frac{\cancel{x^2}\left(4+\cancel{\frac{12}{x}}^{0}+\cancel{\frac{9}{x^2}}^{0}\right)}{\cancel{x^2}\left(-5-\cancel{\frac{2}{x}}_{0}+\cancel{\frac{5}{x^2}}_{0}\right)}$$

$$=\frac{-4}{5}$$

k) $\lim_{x\to 2}\frac{\frac{1}{x-5}+\frac{1}{3}}{x-2}$

$$=\lim_{x\to 2}\frac{3+x-5}{3(x-2)(x-5)}$$

$$=\lim_{x\to 2}\frac{\cancel{x-2}}{3\cancel{(x-2)}(x-5)}$$

$$=\frac{-1}{9}$$

11. The position of a particle moving along the x-axis is given by $s(t) = t^2 - 5t + 4$ where t is the elapsed time in seconds.

(a) What is the position of the particle after 2 seconds?

s(2) = 2^2 – 5(2) + 4 =-2

The particle's position after 2 seconds is 2m to the left of the origin.

(b) Calculate the average velocity of the particle from t = 2s to t = 5s.

$$\begin{aligned} ARoC &= \frac{s(5) - s(2)}{5-2} \\ &= \frac{4-(-2)}{3} \\ &= 2 \text{ m/s} \end{aligned}$$

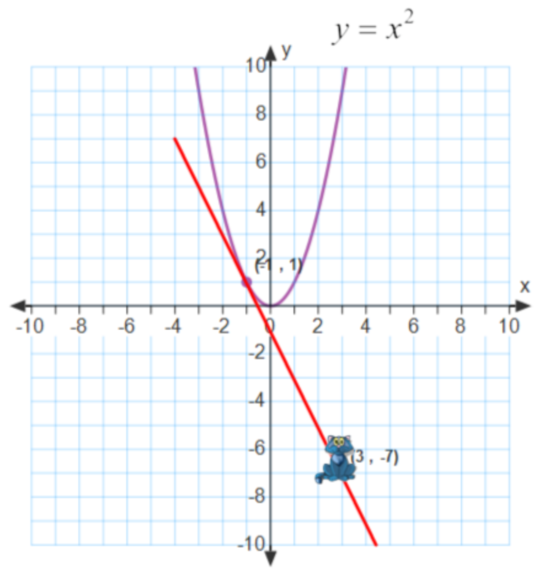
The average velocity is 2m/s.

(c) Calculate the instantaneous velocity of the particle when t = 2s.

$$\begin{aligned} s'(2) &= \lim_{h \to 0} \frac{\left[(2+h)^2 - 5(2+h) + 4\right] - (-2)}{h} \\ &= \lim_{h \to 0} \frac{4 + h^2 + 4h - 10 - 5h + 4 + 2}{h} \\ &= \lim_{h \to 0} \frac{h^2 - h}{h} \\ &= \lim_{h \to 0} \frac{\cancel{h}(h-1)}{\cancel{h}} \\ &= -1 \text{ m/s} \end{aligned}$$

The instantaneous velocity is -1m/s

12. Larry is driving over the speed limit on wet, slippery roads. The road can be modelled by the curve f(x) = x^2. As he travels from left to right, his car begins to slide out of the curve in a straight line at the point (-1, 1). A cat is sitting at the point (3, -7). Will Larry run over the cat?



$$f(x) = x^2$$

$$m = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}, \ a = -1$$

$$m = \lim_{h \to 0} \frac{f(-1+h) - f(-1)}{h}$$

$$y = -2x - 1$$

$$m = \lim_{h \to 0} \frac{(-1+h)^2 - (-1)^2}{h}$$

$$m = \lim_{h \to 0} \frac{1 - 2h + h^2 - 1}{h}$$

$$m = \lim_{h \to 0} \frac{h^2 - 2h}{h}$$

$$m = \lim_{h \to 0} \frac{h(h-2)}{h}$$

$$m = \lim_{h \to 0} (h-2)$$

$$m = -2$$

$$m = -2, \ (-1, 1)$$

$$y = mx + b$$

$$1 = -2(-1) + b$$

$$1 = 2 + b$$

$$-1 = b$$

$$y = -2x - 1$$

$$y = -2(3) - 1$$

$$y = -6 - 1$$

$$y = -7$$

Therefore Larry does run over the cat.

13. Find the point(s) in exact value where the functions $y = 2x^3 + 10x - 1$ and $y = -\frac{4}{x}$ contain the same slope.

$$y = 2x^3 + 10x - 1 \rightarrow y' = 6x^2 + 10$$

$$y = -\frac{4}{x} = -4x^{-1} \rightarrow y' = \frac{4}{x^2}$$

$$6x^2 + 10 = \frac{4}{x^2}$$

$$6x^4 + 10x^2 - 4 = 0$$

$$2\left(3x^4 + 5x^2 - 2\right) = 0$$

$$\left(3x^2 - 1\right)\left(x^2 + 2\right) = 0$$

$$x^2 = \frac{1}{3}$$

$$x = \pm\frac{1}{\sqrt{3}}$$

$$y = \mp 4\sqrt{3}$$

no real roots

14. Determine the values of b and c in the function $f(x) = x^2 - 3bx + (c+2)$, if $f(x)$ has an x-intercept at $x = 1$ and a $f'(3) = 0$.

$$f(x) = x^2 - 3bx + (c+2) \rightarrow f'(x) = 2x - 3b$$

$$f'(3) = 6 - 3b = 0 \rightarrow b = 2$$

$$f(1) = 0 \rightarrow 1^2 - 3(2)(1) + (c+2) = 0$$

$$1 - 6 + c + 2 = 0$$

$$c = 3$$

15. Use the given graph of $f(x)$ to graph $f'(x)$ on the same grid.